Panskura Banamali College (Autonomous)
B.Sc.(H) 5th Semester Examination 2022 (Online mode)
Subject : BCA Paper: CC11T
Paper Name : Analysis of Algorithm & Graph Algorithm
F.M. : 40 Time : 2 Hours

Answer any four questions: **4×10=40**

1. a) What do you mean by worst-case time complexity and best-case time complexity? 4
b) Establish that worst-case time complexity of merge sort is $O(n \times \log_2 n)$. 6

2. a) What do you mean by divide-and-conquer strategy of algorithm design? 3
b) Write an algorithm to merge two sorted sub-arrays $A[1 .. n_1]$, $B[1 .. n_2]$ to produce a sorted array $C[1 .. n_1+n_2]$. 5
c) What do you mean by tail recursion? 2

3. a) Write the recursive algorithm to solve Tower of Hanoi problem with n disks. 5
b) Write an algorithm to find single source shortest path of a graph. 5

4. a) How does greedy method differ from dynamic programming paradigm of algorithm design? 4
b) Write the greedy algorithm to solve knapsack problem. 6

5. a) What do you mean by backtracking algorithm design technique? 4
b) Explain the BFS algorithm to traverse graph vertices 6

6.a. What is Forest?
b. What is Tree?
c. There are five jobs whose profits (P1, P2, P3, P4, P5) = (20, 15, 10, 1, 6) and the deadlines = (2, 2, 1, 3, 3). Find the solution that maximizes profit on scheduling these job.
3+2+5

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